

Operating System

CPU SCHEDULING

CPU Scheduling and Algorithm

Scheduling is the process by which a CPU allocates time to different processes. There are two main types of scheduling: preemptive scheduling and non-preemptive scheduling.

Content includes scheduling types, objectives, CPU and I/O burst cycles, preemptive and non-preemptive scheduling, scheduling criteria, types of scheduling algorithms such as First Come First Served (FCFS), Shortest Job First (SJF), Shortest Remaining Time Next (SRTN), Round Robin (RR), Priority Scheduling, and topics related to deadlock including system models, necessary conditions, prevention, and avoidance.

A scheduling algorithm determines how CPU time is allocated to processes. These can be preemptive or non-preemptive. Preemptive scheduling is based on process priority.

Scheduling is one of the most important functions of the operating system. It ensures maximum utilization of CPU and I/O devices in multiprogramming systems.

CPU Burst and I/O Burst Cycle:

Processes alternate between two states in a cycle — a CPU burst for computations and an I/O burst for input/output operations.

CPU Scheduling Criteria:

1. CPU utilization: Keep the CPU as busy as possible.
2. Throughput: Number of processes completed per unit time.
3. Turnaround time: Total time from process submission to completion.
4. Waiting time: Time spent by a process waiting in the ready queue.
5. Response time: Time taken from process submission to first response.

Types of Scheduling:

Preemptive scheduling allows a process to be interrupted and moved to the ready queue. Non-preemptive scheduling allows a process to hold the CPU until it finishes or enters a waiting state.

Important Concepts:

Arrival Time – time a process arrives in the ready queue.

Completion Time – time a process completes execution.

Burst Time – time required for CPU execution.

Turnaround Time = Completion Time – Arrival Time.

Waiting Time = Turnaround Time – Burst Time.

Gantt Chart – a bar chart used to illustrate scheduling.

Scheduling Algorithms:

1. First Come First Serve (FCFS):

Non-preemptive algorithm. The process that requests first gets the CPU first. Implemented using a FIFO queue. Simple and easy to implement but can cause high waiting times and the convoy effect.

Examples:

Example 1: Burst times 24, 3, 5 → Average waiting time 17, turnaround time 27.6.

Example 2: Burst times 8, 4, 9, 5 → Average waiting time 10.25, turnaround time 16.75.

Example 3: With arrival times 0, 1, 2, 3 and burst times 5, 3, 8, 6 → Average waiting time 5.75, turnaround time 11.25.

Example 4: Burst times 6, 2, 8, 3, 4 and arrival times 2, 5, 1, 0, 4 → Average waiting time 8, turnaround time 11.25 (noted correction).

2. Shortest Job First (SJF):

Selects the process with the smallest burst time next. Reduces average waiting time and is effective for batch processing.

Non-preemptive SJF – process holds CPU until it finishes or waits.

Preemptive SJF – also called Shortest Remaining Time Next (SRTN), where the currently running process can be preempted if a shorter job arrives.

3. Round Robin (RR):

Used in real-time systems. Each process gets CPU for a fixed time quantum in cyclic order. Fair and simple algorithm used widely in operating systems.

Example 1: Burst times 4, 3, 5 → Average waiting time 6.66 (noted mismatch).

Example 2: Burst times 5, 6, 3, 1, 5, 4 and arrival times 0, 1, 2, 3, 4, 6 → Average waiting time 11.83.

Advantages: Prevents starvation, fair CPU sharing, suitable for time-sharing systems.

Disadvantages: More context switching, depends on time quantum, and cannot set priorities.

Deadlock:

Deadlock occurs when two or more processes wait indefinitely for resources held by each other.

Banker's Algorithm (Deadlock Avoidance):

Used to avoid deadlocks by allocating resources safely.

Data Structures:

Available – array showing available resources.

Max – matrix showing maximum demand of each process.

Allocation – matrix showing current resource allocation.

Need – matrix showing remaining needs of each process.

The algorithm consists of two parts: the safety algorithm and the resource request algorithm.

CPU Scheduling in Operating System

MCQ QUESTIONS

1. What is the primary goal of CPU scheduling in an operating system?*

- A) To maximize memory utilization.
 - B) To maximize CPU utilization.
 - C) To minimize the number of context switches.
 - D) To ensure all processes complete their I/O bursts quickly.
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2. The time interval from the time of submission of a process to the time of its completion is called:

- A) Waiting Time
 - B) Response Time
 - C) Turnaround Time
 - D) Throughput
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3. Which of the following is an optimal (but practically impossible to implement) scheduling criterion to minimize the average waiting time?

- A) Maximizing throughput
 - B) Minimizing response time
 - C) Maximizing CPU utilization
 - D) Minimizing turnaround time
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4. A CPU scheduling decision may take place when a process switches from the running state to the ready state. This is an example of:

- A) Non-preemptive scheduling.
- B) Preemptive scheduling.
- C) Co-operative scheduling.
- D) Round-robin scheduling only.

5. Which CPU scheduling algorithm is non-preemptive and implements the principle of a FIFO queue?

- A) Shortest Job First (SJF)
 - B) Round Robin (RR)
 - C) Priority Scheduling
 - D) First-Come, First-Served (FCFS)
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6. The 'Convoy Effect' is a major disadvantage of which scheduling algorithm?

- A) Round Robin (RR)
 - B) Shortest Job First (SJF)
 - C) First-Come, First-Served (FCFS)
 - D) Shortest Remaining Time First (SRTF)
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7. Which scheduling algorithm is provably optimal, providing the minimum average waiting time for a given set of processes?*

- A) Priority Scheduling
 - B) Shortest Job First (SJF)
 - C) Round Robin (RR)
 - D) First-Come, First-Served (FCFS)
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8. The preemptive version of the Shortest Job First (SJF) algorithm is known as:*

- A) Highest Response Ratio Next (HRRN)
 - B) Shortest Remaining Time First (SRTF)
 - C) Priority Scheduling
 - D) Multilevel Queue Scheduling
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9. What is a major drawback of the standard Priority Scheduling algorithm?*

- A) High overhead due to context switching.
- B) The 'Convoy Effect'.

- C) Starvation or indefinite blocking of low-priority processes.
 - D) Difficulty in implementation.
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10. The technique used to prevent starvation in Priority Scheduling by gradually increasing the priority of processes that wait in the ready queue for a long time is called:*

- A) Thrashing
 - B) Compaction
 - C) Aging
 - D) Swapping
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11. Which scheduling algorithm is specifically designed for time-sharing systems to ensure fairness and good response time for interactive processes?*

- A) Shortest Job First (SJF)
 - B) First-Come, First-Served (FCFS)
 - C) Round Robin (RR)
 - D) Priority Scheduling
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12. In Round Robin scheduling, if the time quantum is extremely large (approaching infinity), the algorithm's behavior is equivalent to:*

- A) Shortest Job First (SJF)
 - B) Shortest Remaining Time First (SRTF)
 - C) First-Come, First-Served (FCFS)
 - D) Priority Scheduling
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13. What is the consequence of setting the time quantum too small in the Round Robin algorithm?*

- A) Starvation of processes.
- B) Increase in CPU burst time.
- C) Excessive context switching overhead.

D) Improved throughput.

14. Consider three processes P1, P2, and P3 with burst times 10, 5, and 2, all arriving at time 0. In non-preemptive SJF scheduling, what is the completion time of process P1?*

- A) 2
 - B) 7
 - C) 17
 - D) 10
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15. If all processes have the same priority in a non-preemptive Priority Scheduling algorithm, the algorithm behaves exactly like:*

- A) Shortest Job First (SJF)
- B) Round Robin (RR)
- C) First-Come, First-Served (FCFS)

Provided link for Problem Solving in Artificial Intelligence is:

<https://www.youtube.com/watch?v=yS-TNaJ0Fzs>